

MINISTRY OF INDUSTRY AND MINERAL RESOURCES

MVP1 ADMIN PORTAL FUNCTIONAL SPECIFICATION

Mandatory engines, configuration studios, publishing, versioning, dependency control and runtime-consumption evidence.

Document code	Version	Date
MIM-INSP-MVP1-ADMIN-FS	2.0 - MVP1 Final Documentation Contract	10 July 2026

BINDING DELIVERY RULE

MVP1 is governed by the complete 9-tab, 478-row customer/auditor spreadsheet baseline. No requirement may be silently removed, weakened, generalized away, replaced by mock behavior, or deferred to MVP2. Strategic BRD content is used only to close implementation-critical gaps and strengthen integrity, auditability, security, offline resilience and operability.

Primary audience	Status
Business, Design, Engineering, QA and Fable/Claude implementation teams	Authoritative implementation input - use with the MVP1 acceptance workbook

1. Purpose, authority and scope boundary

This specification is the independently buildable functional contract for the MVP1 Admin Portal. It must be used together with the enhanced MVP1 BRD and the MVP1 Breakdown and Acceptance Contract workbook. The primary functional authority remains all applicable rows from the 478-row Inspection Project.xlsx baseline.

Code	Module	Rows	Mandatory scope
M09	Compliance Configuration	30	Regulations, inspection items, violations, penalties, evidence rules, report packages, dynamic behavior, scoring, triggers, versioning and publishing.

No requirement mapped to this channel may be moved to MVP2, reduced to a static mock, or declared complete without runtime and evidence acceptance.

2. Personas, responsibilities and information architecture

Persona	Primary responsibilities
Compliance Admin	Own regulations, items, violations, penalties, evidence rules, packages and compliance publishing.
Business Admin	Own policy meaning, thresholds, business validation and approval of configuration changes.
Technical Admin	Own workflow, integrations, GIS/geofence, notification/SLA, system-level dependencies and environment-safe configuration.
Reviewer / Approver	Review change impact, validate configuration and approve/publish where separation of duties applies.

Mandatory navigation and screen surfaces

Screen / surface	Purpose and mandatory behavior
Admin home	Clear navigation to every mandatory engine and configuration domain; show draft/published health and unresolved validation.
Regulations library	Hierarchy, applicability, effective dates, source provenance and version history.
Inspection item catalogue	Sections/groups, order, response semantics, required/optional/conditional behavior and guidance.
Violation and penalty configuration	Codes, levels, thresholds, mapping/cardinality, legal basis, penalty ranges and repeat rules.
Evidence rules	Per-item/per-response evidence type, count, size and mandatory rules.
Package and form builder	Compile report/checklist package, sections, items, action forms and output mapping.
Workflow configuration	States, transitions, actors, guards, fields, side effects, SLA, notifications and escalation.
Risk configuration	Variables, sources, weights/formula, thresholds, effective version and explainability.
GIS / geofence configuration	Official locations, polygons/radius, accuracy/arrival thresholds and override governance.
Assignment rules	Automatic/manual methods, skills, workload, capacity, availability, conflict and override rules.
Notification / SLA configuration	Events, recipients, templates, reminders, breach and escalation rules.
Roles and permissions	Role, organizational scope, channel/action/data permission and least privilege.
Publish / version / impact	Validation center, dependency view, impact analysis, simulation/test, approval, publish and audit.

3. Mandatory admin engine operating model

ID	Engine	Admin ownership	Runtime consumers	Mandatory behavior	Acceptance proof	Forbidden shortcut
ENG-01	Regulatory & Compliance Engine	Admin Portal	Web, iPad, Virtual, Review	Manage regulations, item catalogue, applicability, response semantics, dependencies and published runtime versions.	Publish configuration, execute an inspection against it, then prove the submission references the same version	Hard-coded checklist detached from admin configuration

ID	Engine	Admin ownership	Runtime consumers	Mandatory behavior	Acceptance proof	Forbidden shortcut
ENG-02	Dynamic Form & Package Builder	Admin Portal	iPad, Virtual, Review	Build report packages, sections, items, required/optional/conditional behavior, action forms and evidence requirements.	Change a draft rule, publish a new version, prove old inspection remains on old version and new inspection uses new version	Static forms coded screen-by-screen
ENG-03	Workflow Engine	Admin Portal	All channels	States, transitions, allowed actors, guards, required fields, side effects, notifications, SLA and escalation.	Demonstrate configurable transition and prove runtime follows it	Hard-coded status chips called a workflow engine
ENG-04	Risk Engine	Admin Portal	Planning, Factory 360, Operations, Review where applicable	Risk variables, weights/thresholds, source provenance, score calculation, version reference and explainability sufficient for MVP1 targeting/visibility.	Recalculate a factory score from stored inputs and version; show planning/Factory 360 consumption	Decorative risk gauge with no reproducible logic
ENG-05	Assignment Engine	Admin Portal / rules	Planning, Management, Inspector Workspace	Automatic and manual assignment; availability/capacity checks; conflict detection; override traceability.	Publish visits with valid assignments and prove conflict handling	Random assignment or manual dropdown only
ENG-06	GIS, Geofence & Journey Telemetry	Admin + System	Planning, iPad, Operations Center	Official/observed location, geofence, GPS accuracy, journey state, ETA/distance, route deviation, arrival radius, check-in, override and audit.	Complete journey from start to geofence and show synchronized Ops Center evidence	Map widget without governed location events
ENG-07	Evidence & Media Chain of Custody	Admin + System	iPad, Virtual, Review, Factory 360	Evidence capture rules, metadata, hashing/identity, item linkage, time/location/device attribution, upload/sync, immutable submitted reference.	Capture offline and online evidence, sync, review, and prove chain of custody	Loose file upload without item linkage and immutable history
ENG-08	Violation & Penalty Engine	Admin Portal	iPad, Review, Factory 360, Operations	Violation catalogue, severity/level, response-to-violation rules, penalty mapping/cardinality, legal basis and runtime result.	Trigger violation from configured response and prove linked penalty/result/audit	Manual typing of violations/penalties disconnected from configured rules
ENG-09	Review, Return & Version Engine	Admin/Workflow	Review, iPad, Web	Read-only submitted versions; approve/return/reject; immutable reviewer comments; selective unlock; new resubmission version; comparison.	Return two sections, prove all others remain locked, resubmit and compare versions	Editing original submitted version in place
ENG-11	Notification, SLA & Escalation Engine	Admin Portal	All channels	Events, recipients, templates, reminders, SLA timers, overdue rules and escalations required by MVP1.	Demonstrate assignment, return, cancellation, expiry, appointment and overdue notifications	Hard-coded toasts with no event/recipient governance
ENG-12	Audit, Version & Non-Regression Traceability	System / Admin	All channels	Immutable event history, source requirement IDs, configuration versions, actor/time, before/after state, evidence links and acceptance proof.	Trace a source row from spreadsheet to implementation, runtime event, test and evidence	Marking complete without reproducible evidence

4. Admin-native user journeys

SB03 - Compliance Configuration Admin - Rules, Items, Violations and Packages

Regulations, inspection items, violations, penalties, evidence rules, report packages, triggers, versioning and publish.

Step	Configuration slice	Admin action	Output
1	Regulations Library	Create hierarchy, applicability, effective dates and source provenance.	Regulation versions
2	Inspection Item Catalogue	Define items, response types, guidance, order and dependencies.	Item catalogue
3	Violation Mapping	Map responses/items to violation codes, levels and triggers.	Violation matrix
4	Penalty Setup	Map violation outcomes to governed penalties and ranges.	Penalty rules

Step	Configuration slice	Admin action	Output
5	Evidence Rules	Define mandatory/optional evidence types, counts and limits.	Evidence policy
6	Report Package Definition	Compile versioned sections, items, forms and outputs.	Published package
7	Action Form Triggers	Trigger corrective/legal/action forms and owners from results.	Action-form rules
8	Publish & Version Control	Validate dependencies, impact, lock runtime version and audit changes.	Immutable runtime version

SB13 - Compliance Runtime - Package, Response, Evidence, Scoring and Trigger Semantics

Required/optional items, response meanings, evidence override, scoring, violation/penalty mapping and action-form blocking.

Step	Configuration slice	Admin action	Output
1	Resolve Published Package	Runtime selects exact package/version for visit.	Resolved package version
2	Render Sections & Items	Show ordered sections/groups and item semantics.	Dynamic inspection form
3	Required / Optional / Conditional	Apply item-level mandatory rules.	Validation behavior
4	Response Meaning Mapping	Map user response to configured compliance result.	Compliance result
5	Evidence Override Rules	Require evidence based on item and response.	Evidence requirement
6	Scoring & Exclusion	Calculate score and excluded responses/items correctly.	Score result
7	Violation Trigger	Create configured violation from result.	Violation record
8	Penalty Mapping	Resolve governed penalty relationship.	Penalty outcome
9	Action Form Blocking	Open mandatory action forms and block submission if incomplete.	Action-form state
10	Runtime Lock & Audit	Keep active inspection on immutable published version.	Version/audit reference

SB18 - MVP1 Admin Portal Information Architecture

Navigation and ownership for compliance configuration, lookups, permissions, notification/SLA settings and supporting governance.

Step	Configuration slice	Admin action	Output
1	Admin Home & Navigation	Expose all MVP1 configuration domains as governed workspaces.	Admin IA
2	Regulatory Configuration	Regulations, items, responses, violations and penalties.	Compliance config
3	Package & Form Configuration	Packages, sections, items, evidence and action forms.	Runtime package config
4	Workflow Configuration	States, transitions, guards, side effects and SLA.	Workflow config
5	Risk Configuration	Variables, sources, weights, thresholds and explainability.	Risk config
6	GIS / Geofence Configuration	Official locations, polygons/radius and thresholds.	Location config
7	Assignment Rules	Automatic/manual strategies, capacity and conflict policies.	Assignment config
8	Notifications / SLA	Events, recipients, templates, timers and escalation.	Communication config
9	Roles & Permissions	Channel/action/data permissions and organizational scope.	RBAC config
10	Version / Publish / Impact	Validate dependencies, simulate impact, publish, audit and rollback policy.	Published config

SB20 - MVP1 Architecture, Integrations, Security and Traceability Blueprint

Channel architecture, shared services, source systems, audit, offline sync, media, notifications, GIS and non-regression traceability.

Step	Configuration slice	Admin action	Output
1	Identity & Access Layer	Authentication, RBAC and organizational scope.	Trusted access
2	Channel Layer	Admin, Web, iPad, Virtual and Operations Center.	User experiences
3	Domain Services	Planning, factory, inspection, review, compliance and operations.	Business services
4	Configuration Engines	Regulatory, forms/packages, workflow, risk, assignment and SLA.	Configured behavior
5	Field Services	GIS/geofence, journey telemetry, evidence/media and offline sync.	Field execution
6	Version & Audit	Immutable submissions, configuration versions and event history.	Traceability
7	Data & Search	Factory master, operational stores, search and reporting.	Trusted data
8	Integration Layer	Source systems, notifications, identity and external dependencies.	Interoperability
9	Security & Resilience	Encryption, recovery, fault isolation, monitoring and retention.	Operational assurance
10	Acceptance & Regression	Requirement IDs, tests, screenshots, evidence and zero-regression gates.	Delivery proof

5. Hard admin rule: configuration must drive runtime

No engine is accepted because an admin screen exists. Acceptance requires a complete proof chain: draft configuration -> validation -> publish/version -> runtime object resolves the published version -> business behavior changes accordingly -> audit shows the exact configuration/version used. A plain CRUD list disconnected from runtime behavior is a P0 failure.

- Workflow states and transitions must be configurable and runtime actions must be evaluated against the published workflow version.
- Risk must be reproducible from stored inputs, formula/version and drivers; a decorative score is not accepted.
- Compliance packages, response semantics, evidence rules, scoring, violations, penalties and action-form triggers must be published configuration consumed by inspection runtime.
- GIS/geofence thresholds and boundaries must be governed configuration referenced by location/check-in evidence.
- Dependency validation and impact analysis must block unsafe publishing.
- Published configuration referenced by active inspections is immutable; new changes create new effective versions.

6. Cross-cutting controls and non-functional requirements

- Role-based access and least privilege apply to every screen, action and data scope.
- Every state-changing action must be auditable with actor, time, object, before/after state and version references where relevant.
- No submitted inspection, published configuration, reviewer decision or immutable audit record may be edited silently in place.
- Validation errors must preserve user work and identify exact blockers; partial state transitions are prohibited.
- All retryable operations must be idempotent so retries cannot create duplicate visits, evidence, notifications or submissions.
- The UI must use high-contrast, readable typography, strong hierarchy and consistent canonical components across all modules.
- Accessibility, localization readiness and responsive/device-specific behavior must be designed as system qualities rather than late-stage patches.
- Accepted behavior becomes part of the protected regression baseline.

7. Admin UI/UX authority

- Complex engines require control-plane patterns: left hierarchy/tree, central editor or canvas, right inspector/validation panel, persistent draft/published state and explicit impact warnings.
- Do not implement workflow, package, form, risk or regulation administration as a plain table plus edit modal.
- Every dangerous publish/deactivate/change action must expose dependencies, affected runtime objects and confirmation context.
- Use Mobbin only to discover proven interaction patterns; synthesize them into one Inspection Platform design system and reject inconsistent copy-paste assembly.

8. Complete channel requirement catalogue

This appendix preserves all 30 spreadsheet requirement rows mapped to this functional specification. The acceptance workbook remains the machine-readable execution contract.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M09-001	Regulation Management	Create, edit draft, deactivate, view regulation.	Store regulation, status, effective date, attachments, issuing authority.	Regulation is the parent of inspection items. Deactivated regulation prevents future use but does not affect previous inspections.	Admin can create and manage regulation records.
MVP1-M09-002	Inspection Item Management	Create inspection item, link to regulation, define response type, evidence rule.	Store item configuration and make it available for packages.	Item must belong to a regulation. Item can be reused in multiple packages.	Item can be configured and reused.
MVP1-M09-003	Violation Management	Create violation, define severity, corrective action, grace period.	Link violation to inspection item.	Violation is generated automatically during inspection. Inspector cannot override predefined violation.	Violation is triggered correctly.
MVP1-M09-004	Penalty Management	Configure penalty type, amount if applicable, grace/due rules, linked template.	Store one penalty against the violation.	One violation has only one penalty. Penalty is shown to inspector as information and issued only after approval.	Penalty is linked and displayed correctly.
MVP1-M09-005	Evidence Rule Configuration	Select required evidence types: photo, video, document, comment.	Enforce evidence requirements during inspection execution.	Evidence rules are configured per inspection item. Missing mandatory evidence blocks submission.	Evidence requirement is enforced correctly.
MVP1-M09-006	Report Package Management	Create package, select form/report from Template Builder, attach inspection items.	Store package definition and make it selectable during preparation.	Package can contain one or more forms and inspection items. Visit can support one or multiple packages if configured.	Package can be created and used in execution.
MVP1-M09-007	Package Item Assignment	Add/remove items, mark required/optional, define order/category.	Store package-item relationship.	Same item can be reused across packages. Required items cannot be removed during execution. Optional items may be selected/unselected by inspector.	Package checklist renders correctly.
MVP1-M09-008	Dynamic Template Assignment	Select template from Dynamic Form/Template Builder.	Store template reference and version.	Template configuration is managed in Template Builder. Compliance only consumes active templates.	Correct template is used at runtime.
MVP1-M09-009	Action Form Trigger Configuration	Link action form template to violation/penalty where needed.	Generate action form when configured trigger occurs.	Action forms are generated dynamically during inspection and must be completed if mandatory.	Correct action form is generated.
MVP1-M09-010	Configuration Versioning	Publish new version.	Create immutable version of regulation/items/violations/penalties/package.	Execution uses locked published version. Changes affect future inspections only.	Previous inspections remain traceable to old configuration.
MVP1-M09-011	Draft / Publish Lifecycle	Save draft, validate, publish, deactivate.	Validate dependencies before publish.	Only published configurations can be used in planning/preparation. Draft cannot be used in live inspection.	Invalid configuration cannot be published.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M09-012	Dependency Validation	Click Publish.	Validate regulation, items, response types, violations, penalties, evidence rules, templates.	Package cannot be published if required dependencies are missing.	Admin receives clear validation errors.
MVP1-M09-013	Impact Analysis	View affected packages/items/templates.	Display packages, active visits, and future executions affected by the change.	Published changes affect future usage only, not locked inspections.	Admin can understand impact before publishing.
MVP1-M09-014	Deactivation Rules	Deactivate with reason.	Mark config inactive and prevent future use.	Deactivation does not delete historical data. Reason is mandatory.	Deactivated config no longer appears for new packages.
MVP1-M09-015	Audit Trail	None	Log create, update, publish, deactivate, template link, item assignment.	All configuration changes are immutable and auditable.	Full history is available.
MVP1-M09-016	Inspection Section / Group Management	Create section, name AR/EN, define order, assign items.	Store section and render it in execution checklist.	Sections are configurable per package. Items must appear under a section or default group.	Sections appear correctly in execution.
MVP1-M09-017	Item Display Order	Set item order.	Save sort order and render items accordingly.	No hardcoded order. Order follows package configuration.	Items display in configured order.
MVP1-M09-018	Mandatory Rule	Select Required / Optional / Conditional.	Store rule and enforce during execution.	Required items cannot be unselected. Optional items can be unselected. Conditional items appear only when conditions are met.	Execution respects required/optional/conditional behavior.
MVP1-M09-019	Response Mapping	Map each response to compliance result.	Store mapping and use it during inspection execution.	Response outcome must be configurable, not hardcoded.	Selected response triggers correct result.
MVP1-M09-020	Compliance Result Mapping	Define Compliant / Non-Compliant / N/A / Needs Review.	Evaluate response during execution and produce result.	Non-compliant may trigger violation. N/A may be excluded from scoring. Needs Review may be flagged for reviewer.	Correct compliance result appears during inspection.
MVP1-M09-021	Conditional Item Logic	Define condition rules.	Evaluate condition at runtime and show/hide items.	Hidden conditional items are excluded from mandatory validation.	Conditional items appear only when rule is met.
MVP1-M09-022	Conditional Mandatory Logic	Define when item becomes mandatory.	Enforce mandatory rule only when condition applies.	Conditional mandatory items block submission only when active.	Submission validation respects conditional mandatory logic.
MVP1-M09-023	Item Weight / Score	Enter weight/score.	Store score and pass it to health/risk calculation after approval.	Score does not replace violation logic. It supports analytics/health calculation.	Item score is stored and available for calculations.
MVP1-M09-024	Score Exclusion Rule	Enable/disable scoring.	Mark item as excluded from score calculation.	N/A or excluded items do not affect score.	Excluded items are ignored in score.
MVP1-M09-025	Response-Based Evidence Rule	Define evidence rule per response.	Enforce evidence requirement at execution.	Base rule is per item; response-level evidence can override if configured.	Evidence validation applies correctly.
MVP1-M09-026	Violation Trigger Rule	Link non-compliant result to violation.	Generate violation when mapped response is selected.	Inspector cannot override generated violation.	Correct violation is generated.
MVP1-M09-027	Action Form Requirement Rule	Mark action form optional/mandatory.	Enforce completion if mandatory.	Mandatory action forms block submission if incomplete.	Execution blocks submission when required form is incomplete.
MVP1-M09-028	Configuration Preview	Click Preview.	Render package as it will appear to inspector.	Preview does not publish configuration.	Admin can preview checklist and rules.
MVP1-M09-029	Publish Validation	Click Publish.	Validate sections, items, response mapping, violations, penalties, evidence, templates, action forms.	Package cannot be published with missing dependencies.	Admin sees clear validation errors before publish.
MVP1-M09-030	Runtime Locking	None	Lock published package version when selected for visit.	Later configuration changes do not affect active/previous inspections.	Execution uses the locked version only.